

Lumosaic: Hyperspectral Video via Active Illumination and Coded-Exposure Pixels

Dhruv Verma¹ Andrew Qiu¹ Roberto Rangel² Ayandev Barman² Hao Yang² Chenjia Hu²
 Fengqi Zhang² Roman Genov² David B. Lindell¹ Kiriakos N. Kutulakos¹ Alex Mariakakis¹

¹Department of Computer Science, University of Toronto, Canada

²Department of Electrical & Computer Engineering, University of Toronto, Canada

dhruvverma@cs.toronto.edu

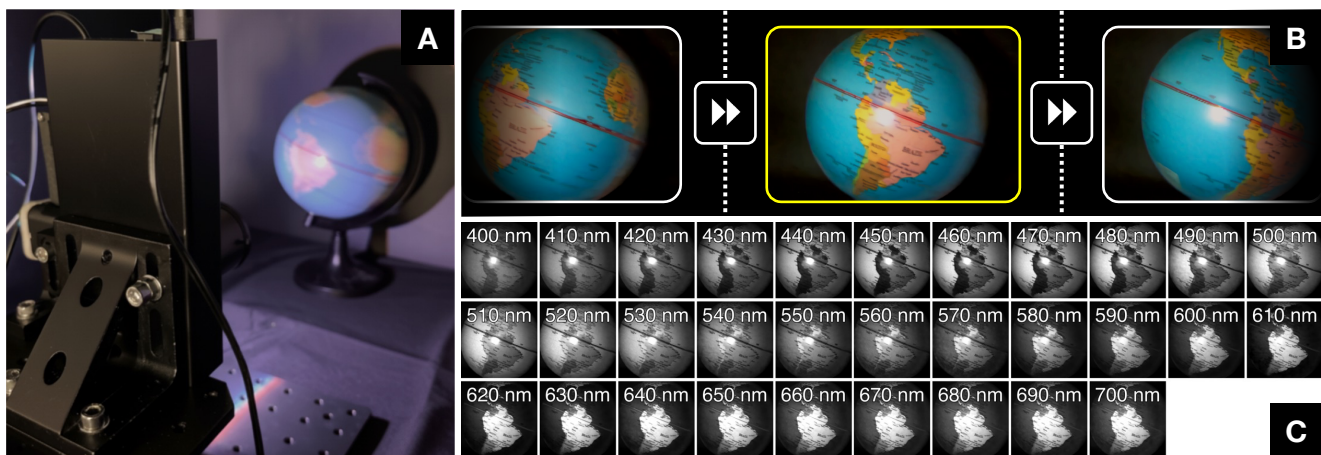


Figure 1. (A) Lumosaic is a compact active hyperspectral video system that combines programmable narrowband illumination with pixel-wise coded exposure to jointly encode spatial, spectral, and temporal information within each video frame. (B) Reconstructed frames of a rotating globe, rendered in sRGB, showing smooth temporal progression despite recording rapid motion at 30 fps. (C) Reconstructed spectral channels (400–700 nm, 10-nm intervals) from the highlighted frame.

Abstract

We present Lumosaic, a compact active hyperspectral video system designed for real-time capture of dynamic scenes. Our approach combines a programmable narrowband LED array with a coded-exposure-pixel (CEP) camera capable of high-speed pixel-wise exposure, thereby enabling joint encoding of scene information across space, time, and wavelength within each video frame. Unlike passive snapshot systems that divide light across multiple spectral channels simultaneously and assume no motion during a frame’s exposure, Lumosaic actively synchronizes illumination and sub-frame exposures to improve photon utilization and preserve spectral fidelity under motion. A learning-based reconstruction pipeline then recovers 31-channel hyperspectral (400–700 nm) video at 30 fps and VGA resolution, producing temporally coherent and spectrally

accurate reconstructions. Experiments on synthetic benchmarks and real-world dynamic scenes demonstrate that Lumosaic significantly improves reconstruction fidelity and temporal stability over existing snapshot hyperspectral imaging systems, enabling robust hyperspectral video across diverse materials and motion conditions.

1. Introduction

Hyperspectral imaging (HSI) captures scene reflectance across several bands of the electromagnetic spectrum. Doing so reveals spectral features invisible to conventional RGB cameras, enabling a wide range of applications such as material classification [48], physiological monitoring [27], spectral relighting [20], and industrial inspection [44]. Despite decades of progress, HSI has yet

Code and data available at <https://github.com/chai-toronto/lumosaic>

to achieve sufficiently high speeds to support dynamic applications with *hyperspectral video*. This challenge is rooted in three compounding constraints: high spectral resolution requires many narrow bands, and narrower bands require longer exposures to account for less light, but video capture requires short exposures. No existing system fully resolves these issues.

Traditional HSI systems rely on spatial scanning that favors spectral fidelity over acquisition speed because it sequentially gathers measurements over multiple seconds. Snapshot HSI methods using coded apertures (e.g., CASSI) [2], diffractive optical elements (DOEs) [4], or multispectral filter arrays (MSFAs) [11] address this limitation by compressing spectral information into a single exposure through static optical encodings. Still, filtering reduces photon efficiency, and simultaneously encoding all spectral channels irreversibly entangles spatial and spectral content during scene motion, thereby degrading reconstruction accuracy.

Active illumination approaches improve photon efficiency by modulating the emitted spectrum rather than filtering it [13, 21, 32, 36, 37, 39]. However, acquiring N spectral bands at video frame rates naively requires an N -fold increase in sensor readout bandwidth, making brute-force hyperspectral video impractical with conventional sensor architectures. Illumination multiplexing [32] and structured light methods [36, 37] reduce the number of captures required, yet a complete set of measurements with several spectral bands still spans multiple frames. This leaves spectral reconstruction vulnerable to the same inter-frame motion artifacts that limit scanning systems.

In a system called *Lumosaic* (see Figure 1), we demonstrate that motion-robust hyperspectral video can be achieved by jointly encoding space, time, and spectra within each video frame. *Lumosaic* combines a programmable narrowband LED array and a cutting-edge *coded-exposure-pixel* (CEP) image sensor [14, 28, 41, 46]. As the system rapidly cycles through multiple LED activations, pixels are exposed in a time-varying mosaic pattern to create a dense *spatio-spectro-temporal encoding* of the scene within a single video frame. *Lumosaic*'s hardware acts as an optical encoder, while a software pipeline decodes the signal through spectral demosaicing, motion alignment via optical flow, and a learning-based reconstruction stage that recovers 31-channel hyperspectral video (400–700 nm) from coded video frames captured at 30 fps. Unlike prior systems that relied on bulky, alignment-sensitive, and aberration-prone optics [10, 25, 38], our signal acquisition operates entirely in silicon, enabling compact and calibration-free deployment. Rather than filtering broadband lighting, *Lumosaic*'s active approach allows each LED's

narrowband output to contribute fully to the captured signal, thereby improving photon efficiency under dynamic or low-light conditions.

Our main technical contributions include:

- *Lumosaic*, a compact active hyperspectral video system that captures 31-channel motion-robust hyperspectral video (400–700 nm) at 30 fps, demonstrated on a hardware prototype integrating a CEP image sensor with a programmable narrowband LED array.
- A jointly designed illumination–exposure coding scheme that tiles 12 narrowband spectral bands and pixel-wise temporal codes across the sensor in a dense spatio-spectro-temporal mosaic, compressing a complete spectral observation into a single video frame without dispersive optics or multi-frame acquisition.
- A learning-based reconstruction pipeline that recovers temporally coherent hyperspectral video at the sensor's native VGA resolution (640×480) from coded frames.

Through extensive experiments on synthetic and real data, we demonstrate that *Lumosaic* achieves higher reconstruction fidelity than state-of-the-art snapshot HSI approaches. We showcase its performance across diverse scenes with varying spatial, spectral, and motion characteristics. These results establish a new sensing paradigm at the intersection of active illumination and coded-exposure imaging for practical hyperspectral videography in dynamic environments.

2. Related Work

Spectral Video Acquisition. The most common HSI paradigm acquires spectral information across multiple exposures, whether by scanning the scene spatially [33] or cycling through narrowband filters [15] or illumination states [13]. These systems achieve high spectral fidelity but require long acquisition times that preclude dynamic scenes. Several efforts have attempted to reduce this bottleneck via multiplexing of spatial, spectral, and temporal signals. Illumination multiplexing [32] captures multiple frames under distinct combinations of illuminants with a global-shutter camera, creating an inherent trade-off between spectral resolution and motion robustness. Meanwhile, structured-light methods [21, 37] project spatially coded patterns across multiple exposures, achieving at most 6.6 fps to date.

Others exploit high temporal sampling to account for scene motion. ChromaFlash [39] compresses spectral acquisition into a single frame by flashing distinct narrowband LEDs as each image row is exposed with a rolling-shutter camera. However, fast-moving objects become skewed because they appear at shifted positions across rows. Yu et al. [45] pair a sweeping-rainbow illumination pattern with an event camera to encode

spectral changes as asynchronous events, but they require mechanically rotating optics and cannot disambiguate spectral responses from scene motion orthogonal to the sweep direction. SpectraTrack [22] augments a line-scan camera with an auxiliary 100-fps RGB sensor to retroactively compensate for inter-line motion via optical flow, but they cannot recover spectral information for objects that enter or exit the scene mid-scan. In summary, all three systems result in poor spectral reconstruction under motion because of their sequential readout geometry or extended acquisition window.

Passive Snapshot Encoding. Snapshot HSI methods encode the full spectral cube into a single exposure through optical elements, eliminating sequential acquisition entirely. Early approaches such as CASSI [2, 12, 40] employed dispersive optics, coded apertures, and relay lenses to achieve single-shot acquisition at the expense of bulky equipment, extensive calibration, and complex inversion. Subsequent designs improved compactness yet introduced new limitations: prism-based systems [3] suffer from chromatic distortion and spatial blur due to dispersion; printed color-dot masks [47] have low optical throughput and ill-conditioned spectral responses; and MSFAs [11] capture very little light due to absorptive filtering, attenuating effective signal contribution under time-constrained scenarios such as motion and video. DOEs [4, 23, 35] and metasurface approaches [8, 29] offer high spectral diversity in miniature form factors but remain sensitive to fabrication imperfections, wavelength-dependent efficiency, and static spatial encoding that breaks under high-speed motion or noise.

A common limitation emerges across passive snapshot designs: encoding all spectral channels simultaneously requires dividing available photons across bands within a single exposure. At short exposures, per-channel SNR drops below reconstruction thresholds; at long exposures, scene motion irreversibly entangles spatial and spectral content during integration. No static optical design can circumvent this trade-off, rendering passive snapshot systems incompatible with the photon efficiency and motion robustness that hyperspectral video demands.

Coded-Exposure-Pixel Imaging. Pixel-wise coded exposure was first demonstrated for compressive video acquisition by Hitomi et al. [16], who showed that temporally modulating pixels enables high-speed video recovery from a single coded frame. Llull et al. [26] subsequently showed that varying coded apertures could compress multiple temporal frames into a single snapshot, establishing the broader paradigm of snapshot compressive temporal imaging. Modern coded-exposure-pixel sensors, also known as multi-tap or multi-bucket sensors [6, 14, 19, 28, 41], realize this capability directly in

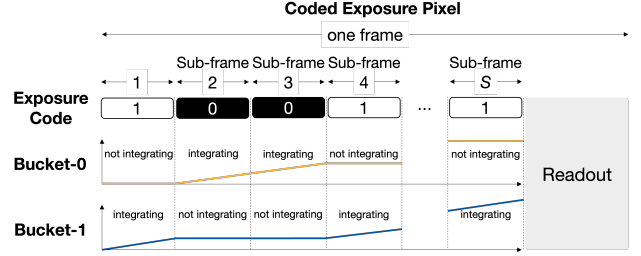


Figure 2. The operation of a single pixel in a coded-exposure-pixel (CEP) camera. Each pixel alternates between two charge storage sites (Bucket 0 and Bucket 1) according to a binary exposure code. During each sub-frame, only one bucket integrates incident light, enabling temporal multiplexing across the exposure period.

hardware at rates exceeding 39 kHz [14], enabling dense spatio-temporal coding without external optical modulation that earlier coded-exposure designs required. Although coded-exposure imaging has been explored for motion deblurring [34], HDR imaging [30], and transient analysis [25], its potential for hyperspectral video remains largely untapped. The key insight for Lumosaic is that a CEP camera can associate each pixel with a narrow spectral band using time-varying illumination. When the pixel assignments are tiled in a dense mosaic over several measurements, the readout geometry does not induce significant distortions under motion and hyperspectral video can be captured at 30 fps. Using an active approach also allows each illuminant to contribute its full photon budget to the signal rather than filtering it away.

3. Imaging Forward Model

This section describes the imaging principles, coding scheme, and measurement model underlying Lumosaic.

3.1. Coded-Exposure-Pixel Camera

A CEP camera differs from a conventional global-shutter image sensor in two key features. First, each pixel of the CEP camera incorporates two charge-collection sites, or *buckets*, enabling the segregation of photo-induced charge accumulation. Second, the exposure duration of each pixel within a video frame is divided into smaller intervals called *sub-frames*. Each pixel includes a one-bit writable memory that facilitates programmable control over which bucket is active during each sub-frame.

Figure 2 illustrates the operation of a CEP camera pixel and the corresponding integration process over sub-frames. Charge accumulation occurs at two different timescales:

1. **Sub-frame level:** The charge collected at each sub-frame is accumulated and transferred to the active bucket based on each pixel's control signal, and
2. **Frame level:** The total charge accumulated across all sub-frames is integrated and read out for each bucket.

Programming a CEP camera involves specifying the number of sub-frames per video frame, the duration of each sub-frame, and the state of each pixel's exposure at each sub-frame. The exposure schedule is represented by a binary matrix $\mathbf{C} \in \{0, 1\}^{P \times S}$, where P is the number of pixels and S is the number of sub-frames; $C_{p,s} = 1$ indicates that pixel p is active during sub-frame s .

3.2. Time-Varying Spectral Illumination

We deliver spectral modulation by using an array of L narrowband LEDs, each with a distinct spectral power distribution $E_l(\lambda)$, where λ denotes wavelength. The illumination is varied in synchrony with the sub-frame exposure schedule of the CEP camera.

We define the illumination schedule using a binary matrix $\mathbf{I} \in \{0, 1\}^{S \times L}$, where $I_{s,l} = 1$ indicates that LED l is active during sub-frame s . The resulting spectral irradiance incident on the scene for a given sub-frame is

$$\mathcal{I}_s(\lambda) = \sum_{l=1}^L I_{s,l} \cdot E_l(\lambda). \quad (1)$$

This temporally varying illumination encodes spectral content directly into the exposure sequence. Together, the illumination matrix $\mathbf{I}$ and the exposure code matrix $\mathbf{C}$ form the basis for our measurement model.

3.3. Joint Illumination and Exposure Coding

To enable dense and structured multiplexing, we design a coding scheme that synchronizes LED activations with pixel exposures across tightly packed neighborhoods on the sensor. As shown in Figure 3 (B), we cycle through L narrowband LEDs across S sub-frames while spatially tiling both illumination and exposure schedules across the sensor in a mosaic-like fashion. Pixels are grouped into T repeating tiles (e.g., a 4×4 pixel mosaic yields $T = 16$ tiles) with unique exposure schedules.

We define tile exposure and illumination codes as

$$\mathbf{C}_{\text{tile}} \in \{0, 1\}^{T \times S}, \quad \mathbf{I}_{\text{tile}} \in \{0, 1\}^{T \times S \times L}. \quad (2)$$

Here, $\mathbf{C}_{\text{tile}}[t, s] = 1$ indicates that pixels in tile t are actively integrating light during sub-frame s , and $\mathbf{I}_{\text{tile}}[t, s, l] = 1$ specifies that LED l is turned on during sub-frame s for tile t . We define a mapping function

$$\pi : \{1, \dots, P\} \rightarrow \{1, \dots, T\} \quad (3)$$

that assigns each pixel index p to its corresponding tile index $\pi(p)$. Using this mapping, each pixel inherits its exposure and illumination codes as

$$C_{p,s} = \mathbf{C}_{\text{tile}}[\pi(p), s], \quad I_{p,s,l} = \mathbf{I}_{\text{tile}}[\pi(p), s, l]. \quad (4)$$

Thus, all pixels with the same tile index share the same illumination pattern, while adjacent pixels observe different wavelength bands at different times. Over S sub-frames, this coordinated scheme produces a temporally staggered *spectral mosaic*—a single video frame in which

different spatial locations encode distinct spectral and temporal samples of the scene.

3.4. Measurement Model

The goal of our measurement model is to represent each hyperspectral video frame as a matrix $\mathbf{R} \in \mathbb{R}^{P \times \Lambda}$, where each row $\mathbf{r}_p \in \mathbb{R}^\Lambda$ describes the spectral reflectance at a pixel p according to Λ discrete wavelength bands.

From Equation 1, the time-varying illumination at sub-frame s and pixel p can be expressed discretely as

$$\mathcal{I}_{p,s} = \sum_{l=1}^L I_{p,s,l} \mathbf{E}_l, \quad (5)$$

where $\mathbf{E}_l \in \mathbb{R}^\Lambda$ denotes the spectral power distribution of LED l , and $I_{p,s,l} \in \{0, 1\}$ specifies whether that LED is active for pixel p during sub-frame s according to its tile schedule. Given the camera's spectral sensitivity $\mathcal{S} \in \mathbb{R}^\Lambda$, the effective spectral sensing vector is

$$\mathbf{a}_{p,s} = \mathcal{S} \odot \mathcal{I}_{p,s}, \quad (6)$$

where $\odot$ denotes element-wise multiplication across the Λ wavelength bands. The photo-response for pixel p in sub-frame s is then

$$y_{p,s} = \mathbf{a}_{p,s}^\top \mathbf{r}_p = \sum_{k=1}^\Lambda \mathcal{S}_k \mathcal{I}_{p,s,k} r_{p,k}. \quad (7)$$

Although the CEP camera supports dual-bucket readout, we discard measurements from Bucket 0 and retain only the integration from Bucket 1 in our implementation. The total measured intensity at pixel p over a frame is obtained by summing all sub-frames in which it is active:

$$Y_p = \sum_{s=1}^S C_{p,s} y_{p,s} + \eta_p, \quad (8)$$

where $C_{p,s} \in \{0, 1\}$ indicates whether pixel p integrates light during sub-frame s , and η_p accounts for sensor noise and other residual errors.

Stacking all pixel measurements into a vector $\mathbf{Y} \in \mathbb{R}^P$ and vectorizing the hyperspectral cube as $\mathbf{x} = \text{vec}(\mathbf{R}) \in \mathbb{R}^{P\Lambda}$ gives the global forward model:

$$\mathbf{Y} = \mathbf{A} \mathbf{x} + \boldsymbol{\eta}, \quad (9)$$

where $\mathbf{A} \in \mathbb{R}^{P \times P\Lambda}$ encodes the combined effects of illumination modulation, exposure coding, and sensor spectral sensitivity. Each row of $\mathbf{A}$ represents the effective spectral integration weights for one pixel aggregated over all active sub-frames. This linear model defines how spatial, spectral, and temporal information are jointly encoded in a single captured frame, forming the basis for our hyperspectral reconstruction pipeline.

3.5. Hardware Prototype

Lumosaic consists of a custom active illumination module and a prototype CEP camera (see Figure 3 (A)).

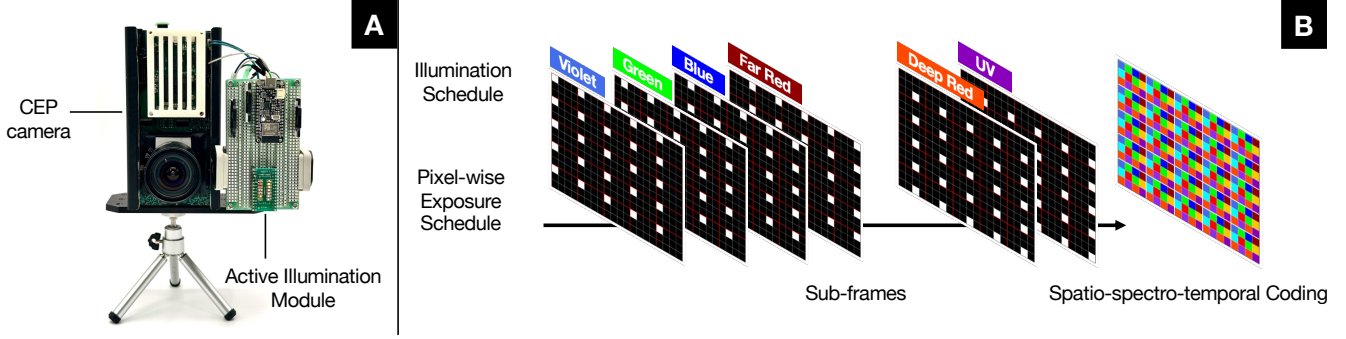


Figure 3. (A) Lumosaic features a CEP camera synchronized with a programmable array of narrowband LEDs. (B) Each sub-frame is assigned a unique LED and spatial exposure mask, producing a dense spatio-spectro-temporal scene encoding within a single video frame.

Active Illumination Module. The illumination module comprises 12 high-power narrowband LEDs (Lumileds Luxeon C), each covering a distinct portion of the visible spectrum with a full-width-at-half-maximum (FWHM) of approximately 20–30 nm. All LEDs are driven by a custom current driver capable of switching at frequencies exceeding 100 kHz. A microcontroller (Adafruit ESP32 Feather v2) generates control signals that synchronize LED activation with the CEP camera’s sub-frame clock, achieving microsecond-level timing precision.

CEP Camera. The prototype uses a VGA-resolution (640×480) CEP image sensor that supports pixel-wise binary exposure modulation with up to 12,500 sub-frames per second. For our implementation, a single video frame comprises $S = 158$ sub-frames, with each sub-frame lasting $150 \mu\text{s}$. This configuration yields a total integration period of $\sim 24 \text{ ms}$ that is suitable for 30-fps video. The primary limiting factor in pushing the frame rate further is not the exposure duty cycle but rather the additional $\sim 6 \text{ ms}$ of pixel readout time required between consecutive frames.

Synchronization and Control. LED activation and sensor exposure schedules are jointly programmed and triggered via hardware-level synchronization lines. Each LED’s activation window is mapped to a contiguous sequence of sub-frames that share a common exposure pattern. To compensate for variations in LED output power and the camera’s spectral sensitivity, LEDs with lower radiance are allocated proportionally longer activation durations, ensuring balanced spectral energy delivery across the frame. No two LEDs are active simultaneously to minimize crosstalk and improve spectral separability.

Calibration. We perform a one-time calibration procedure to characterize (1) the spectral power distribution of the illumination module, (2) the spectral sensitivity of the CEP camera, and (3) the system’s overall radiometric response. These measurements define the parameters of the imaging forward model while ensuring that reconstructed hyperspectral data is both radiometrically accurate and

physically consistent with real-world measurements.

Each of the 12 LEDs is sequentially activated so that its emission spectrum $\mathbf{E}_l$ can be measured with a calibrated spectroradiometer (Konica Minolta CS-2000), providing an accurate spectral power distribution from 380–780 nm. The camera’s spectral sensitivity $\mathcal{S}$ is measured using a monochromator (Image Engineering camSPECS XL) with 5-nm interference filters; the outputs are aggregated into 10-nm bins for alignment with the reconstruction model. To ensure accurate radiometric scaling, we capture a Macbeth ColorChecker and jointly optimize LED-specific gains to equalize integrated intensity across channels, compensating for residual non-uniformities in LED brightness, optical coupling, and sensor response. We use the resulting calibration parameters in both simulation and real-world reconstruction. Detailed procedures and plots are provided in Supplementary Section 1.

4. Hyperspectral Video Reconstruction

Given a coded measurement $\mathbf{Y}_i$ corresponding to frame i , our goal is to reconstruct the underlying hyperspectral scene $\mathbf{R}_i \in \mathbb{R}^{P \times \Lambda}$, where each pixel’s spectral reflectance $\mathbf{r}_p$ spans $\Lambda = 31$ wavelength bands. We first demosaic $\mathbf{Y}_i$ into a set of LED-specific sub-images $\{\mathbf{Y}_i^{(1)}, \mathbf{Y}_i^{(2)}, \dots, \mathbf{Y}_i^{(L)}\}$, where $L = 12$ denotes the number of narrowband LEDs. Each $\mathbf{Y}_i^{(l)}$ aggregates pixels assigned to LED l according to the known illumination–exposure mosaic described in Section 3. We then perform bilinear interpolation to upsample each sub-image to the full spatial resolution of the sensor.

Unlike with passive filter array systems, our sub-images correspond to different time intervals within the exposure period, meaning they may be spatially misaligned in scenes with motion. To avoid artifacts during reconstruction, we incorporate a temporal alignment step beforehand.

4.1. Temporal Alignment of Spectral Channels

Each LED-specific sub-image $\mathbf{Y}_i^{(l)}$ captures scene content under a distinct narrowband illumination. This fact violates

the photometric consistency assumption underlying conventional optical flow methods, making direct alignment between sub-images unreliable. Therefore, we perform temporal alignment by estimating motion across sub-images assigned to the same LED in adjacent frames.

We designate the sub-image illuminated by the lime-coloured LED, $\mathbf{Y}_i^{(\text{lime})}$, as the temporal reference for frame i due to its central wavelength and mid-exposure timing. For each other LED l , we determine its relative position in the illumination schedule and pair its sub-image $\mathbf{Y}_i^{(l)}$ with a temporally adjacent frame. If l occurs before the lime LED in the illumination cycle, we pair it with $\mathbf{Y}_{i+1}^{(l)}$; if it occurs after, we pair it with $\mathbf{Y}_{i-1}^{(l)}$. Using these pairs, we estimate motion fields with the Real-Time Intermediate Flow Estimation (RIFE) network [17], which predicts an interpolated frame at a specified timestamp. We warp each sub-image $\mathbf{Y}_i^{(l)}$ to the reference time of $\mathbf{Y}_i^{(\text{lime})}$, producing the temporally aligned result $\hat{\mathbf{Y}}_i^{(l)}$.

4.2. Learning-based Reconstruction

For each video frame i , the temporally aligned sub-images $\{\hat{\mathbf{Y}}_i^{(1)}, \hat{\mathbf{Y}}_i^{(2)}, \dots, \hat{\mathbf{Y}}_i^{(L)}\}$ serve as input to a deep neural network that reconstructs the corresponding hyperspectral image $\mathbf{R}_i$ with $\Lambda = 31$ spectral channels.

Model Architecture. We adopt the Holistic Attention Network (HAN) [31] as our reconstruction backbone because of its demonstrated utility in image restoration tasks. The model takes a $66 \times 64 \times 12$ tensor as input, corresponding to a spatial patch of the 12 demosaiced and temporally aligned LED sub-images. The model outputs a $66 \times 64 \times 33$ hyperspectral cube spanning 33 spectral bands from 380–780 nm. Two extrapolated edge channels covering the UV (1st) and NIR (33rd) boundaries are included during training to stabilize spectral boundary reconstruction but are discarded at inference. This leaves 31 channels between 400 and 700 nm at 10-nm intervals in the final output. We reconstruct full-frame hyperspectral images in a patch-wise manner and merge them via weighted aggregation to ensure spatial continuity. Architecture details, patch reconstruction, and aggregation procedures are provided in Supplementary Section 3.

Training Setup. We train the reconstruction model by simulating Lumosaic’s forward model on HSI datasets. We emulate real-world capture by adding zero-mean Gaussian noise with a standard deviation uniformly sampled between 0% and 15% of the maximum signal intensity.

For data augmentation, we extract random spatial patches from hyperspectral images and apply random horizontal and vertical flips. Since most public datasets provide spectral measurements between 400–700 nm, we extend the range to 380–780 nm by mirroring the edge channels: the 420-nm and 410-nm bands approximate the

ultraviolet (380–390 nm) region, while channels beyond 710 nm mirror those opposite of 700 nm. Extrapolated channels are used only during training to stabilize spectral boundary reconstruction and are omitted at inference.

We minimize the $\mathcal{L}_1$ loss between the predicted and ground-truth hyperspectral cubes using the Adam optimizer with a learning rate of 1×10^{-4} and default β parameters. We train the model with a batch size of 14 and gradient accumulation over two steps for memory efficiency. Our PyTorch implementation takes approximately 24 hours to go through 50,000 training iterations on an NVIDIA RTX A6000 GPU, and inference on a single 640×480 frame requires 4.7 s.

5. Experiments

We evaluate Lumosaic across simulations and real-world captures, progressing from controlled static reconstructions to dynamic video demonstrations.

5.1. Simulations

Setup. Since publicly available hyperspectral video datasets are limited, our simulations focus on static scene reconstruction. We simulate image formation under matched conditions on a unified corpus of three HSI datasets: CAVE [43] (32 indoor scenes), KAUST [24] (409 indoor and outdoor scenes), and ARAD [1] (949 indoor and outdoor scenes). After resampling each hyperspectral cube to 31 uniformly-spaced channels between 400–700 nm, we divide the corpus into 80%-10%-10% splits for training, validation, and testing, respectively.

We compare reconstruction performance across seven configurations: (1) Lumosaic’s forward model with HAN [31] as the reconstruction backbone; (2) Lumosaic with an MCAN [9] backbone; (3) Lumosaic with an SRNet [5] backbone; (4) Lumosaic with a SPECAT [42] backbone; (5) optical filtering-based compressive HSI paired with the SPECAT [42] reconstruction backbone; (6) QDO [23], an end-to-end optimized DOE-based snapshot HSI system; and (7) MST++ [7], a data-driven RGB-to-hyperspectral reconstruction method.

We train all models from scratch using identical data splits, input normalization, and optimization schedules. We evaluate performance on the held-out test set using standard spatial and spectral quality metrics: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), mean absolute error (MAE), and spectral angle mapper (SAM). Higher PSNR/SSIM and lower MAE/SAM indicate better reconstruction quality.

Noise Robustness. To assess stability under photon-limited conditions typical in fast motion and video capture, we add Gaussian noise levels of $\sigma = \{0\%, 5\%, 10\%, 15\%, 20\%\}$ relative to the maximum

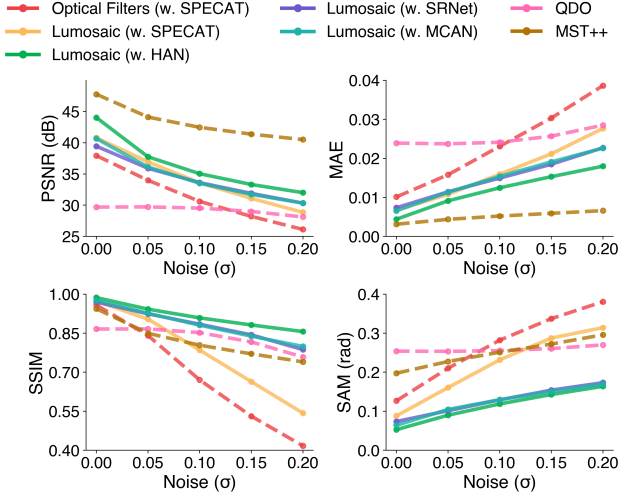


Figure 4. Reconstruction performance of QDO [23], MST++ [7], and Lumosaic using SPECAT [42], HAN [31], SRNet [5], and MCAN [9] backbones according to (top-left) PSNR, (top-right) MAE, (bottom-left) SSIM, and (bottom-right) SAM for the simulations under varying Gaussian noise levels.

signal intensity. As shown in Figure 4, Lumosaic maintains high PSNR while consistently achieving higher SSIM and lower SAM compared to baselines across all noise levels. Additional qualitative results (see Supplementary Section 4) further illustrate that Lumosaic preserves spatial details and accurate spectral reflectance, whereas QDO and MST++ exhibit noticeable blurring and spectral distortion under higher noise.

Reconstruction Backbone Sensitivity. All four Lumosaic-based models spectrally outperform alternative baselines (optical filtering, QDO, and MST++) while maintaining high spatial fidelity, highlighting performance gains attributable to our sensing model rather than to mere network complexity. Among the reconstruction backbones, HAN delivers the best fidelity, reaching 44.0 dB in the noise-free case and 32.0 dB even at $\sigma = 20\%$ (see Figure 4). MCAN and SRNet perform slightly worse than HAN in reconstruction accuracy but require significantly less compute and inference time (52 ms and 27 ms per frame, respectively), making them attractive for real-time deployment. SPECAT achieves comparable performance in low-noise regimes but degrades more noticeably under higher noise levels.

Spectral Resolution. We further evaluate the ability of Lumosaic to recover narrowband spectral features using controlled synthetic scenes. Despite the limited spectral sampling of the illumination hardware, the reconstruction accurately captures sharp spectral transitions within the physical bandwidth and sampling constraints. Detailed

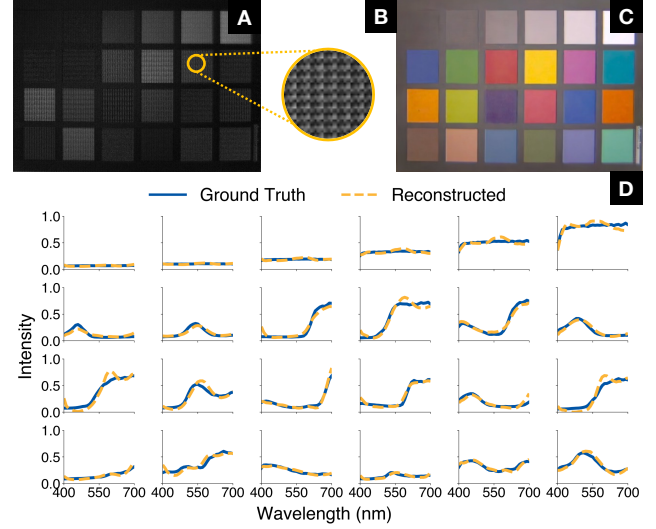


Figure 5. Quantitative and qualitative evaluation of spectral reconstruction accuracy using a real-world ColorChecker target. (A) Raw coded frame captured by Lumosaic. (B) Zoomed-in region showing the mosaic coding. (C) Reconstructed hyperspectral image rendered in sRGB. (D) Spectral reflectance curves for all 24 patches, comparing ground-truth (solid blue) and reconstructed (dashed yellow) spectra.

experiments and visualizations are provided in Supplementary Section 4.3.

5.2. Real-World Evaluation

Static Scenes. To assess spectral and perceptual fidelity, we capture several static scenes: a ColorChecker, optical filters, and everyday objects (e.g., fabrics and figurines). As shown in Figure 5, the reconstructed reflectance spectra of the ColorChecker closely match the spectroradiometer measurements (Konica Minolta CS-2000), validating Lumosaic’s radiometric calibration and spectral reconstruction accuracy. Additional examples in Figure 6 and Supplementary Sections 5.1, 5.2 show faithful color reconstruction, successful disambiguation of metamers (e.g., real vs. fake objects), and high perceptual consistency across diverse materials and textures.

Dynamic Scenes. We evaluate Lumosaic on dynamic scenes exhibiting both rigid (translation, rotation, panning) and non-rigid (hand gestures, liquid diffusion, effervescence) motions. Although millisecond-level offsets between LED sub-images would typically induce motion blur, Lumosaic reconstructs temporally coherent hyperspectral video with high spectral fidelity and stability at 30 fps (see Figure 1 and 7). Figure 8 further illustrates the importance of our flow-based temporal alignment; without it, reconstructions exhibit visible artifacts and boundary distortions, particularly under fast motion. While Lumosaic handles diverse motion types robustly, artifacts

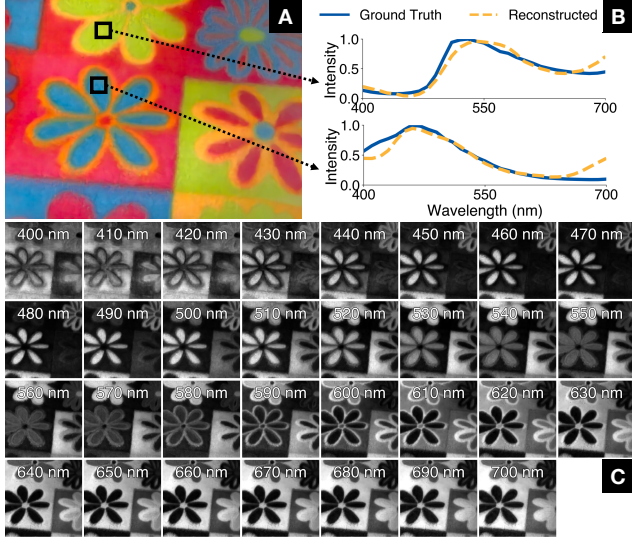


Figure 6. An example of real-world reconstruction results obtained using Lumosaic. (A) Reconstructed hyperspectral image rendered in sRGB. (B) Reflectance spectra extracted from marked regions. (C) Reconstructed hyperspectral channels.

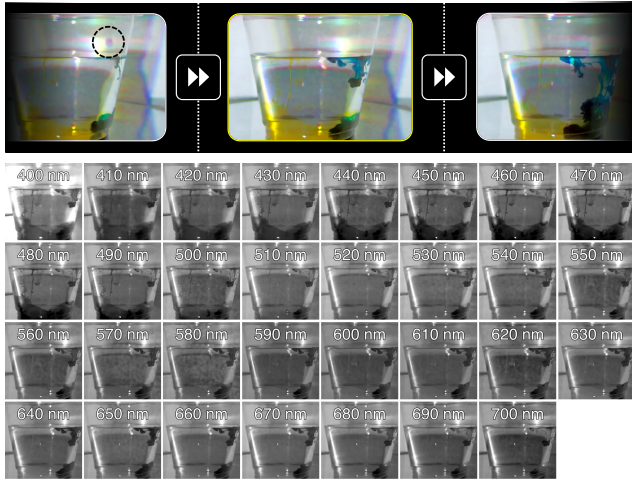


Figure 7. Hyperspectral video reconstruction of a colored droplet diffusing in water, captured using Lumosaic at 30 fps. Each column shows a representative frame over time rendered in sRGB, while the bottom row visualizes reconstructed spectral channels for the highlighted frame. See the Supplementary Video for the full sequence visualization.

occasionally persist at fast, high-contrast boundaries due to two factors: slightly misaligned shadows due to LED placement geometry, and residual flow errors in occluded or motion-blurred regions. Supplementary Section 5.3 and the Supplementary Video present additional results with minimal ghosting or flicker across all motion types.

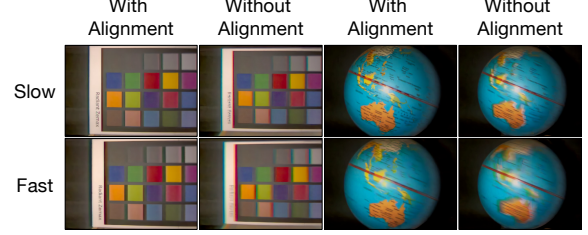


Figure 8. The effect of temporal alignment on Lumosaic’s reconstruction quality in dynamic scenes. We compare hyperspectral reconstructions rendered in sRGB with and without temporal alignment for (top) slow and (bottom) fast motions.

6. Future Work & Concluding Remarks

We have demonstrated that Lumosaic can reconstruct 30-fps hyperspectral video at VGA resolution with 31 spectral channels (400–700 nm). This was made possible by coordinating time-varying illumination with a CEP camera to generate a dense spatio-spectro-temporal encoding within each video frame. Lumosaic’s active, co-designed sensing strategy simultaneously acquires motion and spectral information with high light efficiency—a capability not previously realized in such a compact form factor. Our results show temporally coherent and spectrally faithful reconstructions across diverse materials and motion patterns, bridging the long-standing gap between snapshot imaging and hyperspectral video.

There are several opportunities for future investigation. First, our reconstruction pipeline processes each frame independently—a deliberate design choice that simplifies training given the scarcity of large hyperspectral video datasets, but one that leaves inter-frame temporal priors unexploited. Simulating motion in widely available HSI datasets represents a natural path toward video-aware reconstruction. Second, our implementation only uses measurements from one of the CEP sensor’s two buckets. Jointly modeling both buckets could improve dynamic range, light efficiency, and motion robustness; however, maximizing SNR in the primary bucket currently saturates the other bucket in bright regions, requiring per-bucket gain control in future sensor iterations. Finally, we did not fully explore the trade-offs among different coding designs, as using adaptive or randomized mosaics may offer distinct advantages.

In summary, Lumosaic offers a novel design space for computational hyperspectral video by coupling active illumination with coded-exposure imaging. We envision this framework to enable new opportunities for spectral sensing in robotics, medical imaging, and computational photography.

Acknowledgments

AM, DBL, and KNK acknowledge support from the Natural Sciences and Engineering Research Council of Canada (NSERC) under the RGPIN program. DBL also acknowledges support from the Canada Foundation for Innovation and the Ontario Research Fund. The authors gratefully acknowledge Michael S. Brown (York University) for providing access to a spectroradiometer and for his valuable consultation, and Sejal Bhalla (University of Toronto) for her dedicated efforts in preparing the supplementary video.

References

- [1] Boaz Arad, Radu Timofte, Rony Yahel, Nimrod Morag, Amir Bernat, Yuanhao Cai, Jing Lin, Zudi Lin, Haoqian Wang, Yulun Zhang, Hanspeter Pfister, Luc Van Gool, Shuai Liu, Yongqiang Li, Chaoyu Feng, Lei Lei, Jiaojiao Li, Songcheng Du, Chaoxiong Wu, Yihong Leng, Rui Song, Mingwei Zhang, Chongxing Song, Shuyi Zhao, Zhiqiang Lang, Wei Wei, Lei Zhang, Renwei Dian, Tianci Shan, Anjing Guo, Chengguo Feng, Jinyang Liu, Mirko Agarla, Simone Bianco, Marco Buzzelli, Luigi Celona, Raimondo Schettini, Jiang He, Yi Xiao, Jiajun Xiao, Qiangqiang Yuan, Jie Li, Liangpei Zhang, Taesung Kwon, Dohoon Ryu, Hyokyoung Bae, Hao-Hsiang Yang, Hua-En Chang, Zhi-Kai Huang, Wei-Ting Chen, Sy-Yen Kuo, Junyu Chen, Haiwei Li, Song Liu, Sabarinathan, K Uma, B Sathya Bama, and S. Mohamed Mansoor Roomi. Ntire 2022 spectral recovery challenge and data set. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 863–881, 2022. [6](#), [4](#)
- [2] Gonzalo R Arce, David J Brady, Lawrence Carin, Henry Arguello, and David S Kittle. Compressive coded aperture spectral imaging: An introduction. *IEEE Signal Processing Magazine*, 31(1):105–115, 2013. [2](#), [3](#)
- [3] Seung-Hwan Baek, Incheol Kim, Diego Gutierrez, and Min H Kim. Compact single-shot hyperspectral imaging using a prism. *ACM Transactions on Graphics (TOG)*, 36(6):1–12, 2017. [3](#)
- [4] Seung-Hwan Baek, Hayato Ikoma, Daniel S Jeon, Yuqi Li, Wolfgang Heidrich, Gordon Wetzstein, and Min H Kim. Single-shot hyperspectral-depth imaging with learned diffractive optics. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2651–2660, 2021. [2](#), [3](#)
- [5] Liheng Bian, Zhen Wang, Yuzhe Zhang, Lianjie Li, Yinuo Zhang, Chen Yang, Wen Fang, Jiajun Zhao, Chunli Zhu, Qinghao Meng, et al. A broadband hyperspectral image sensor with high spatio-temporal resolution. *Nature*, 635(8037):73–81, 2024. [6](#), [7](#)
- [6] Gil Bub, Matthias Tecza, Michiel Helmes, Peter Lee, and Peter Kohl. Temporal pixel multiplexing for simultaneous high-speed, high-resolution imaging. *Nature methods*, 7(3): 209–211, 2010. [3](#)
- [7] Yuanhao Cai, Jing Lin, Zudi Lin, Haoqian Wang, Yulun Zhang, Hanspeter Pfister, Radu Timofte, and Luc Van Gool. Mst++: Multi-stage spectral-wise transformer for efficient spectral reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 745–755, 2022. [6](#), [7](#), [4](#), [5](#)
- [8] MohammadSadeh Faraji-Dana, Ehsan Arbabi, Hyounghan Kwon, Seyedeh Mahsa Kamali, Amir Arbabi, John G Bartholomew, and Andrei Faraon. Hyperspectral imager with folded metasurface optics. *Acs Photonics*, 6(8):2161–2167, 2019. [3](#)
- [9] Kai Feng, Yongqiang Zhao, Jonathan Cheung-Wai Chan, Seong G Kong, Xun Zhang, and Binglu Wang. Mosaic convolution-attention network for demosaicing multispectral filter array images. *IEEE Transactions on Computational Imaging*, 7:864–878, 2021. [6](#), [7](#)
- [10] Wei Feng, Fumin Zhang, Xinghua Qu, and Shiwei Zheng. Per-pixel coded exposure for high-speed and high-resolution imaging using a digital micromirror device camera. *Sensors*, 16(3):331, 2016. [2](#)
- [11] Bert Geelen, Nicolaas Tack, and Andy Lambrechts. A compact snapshot multispectral imager with a monolithically integrated per-pixel filter mosaic. In *Advanced fabrication technologies for micro/nano optics and photonics VII*, pages 80–87. SPIE, 2014. [2](#), [3](#)
- [12] Michael E Gehm, Renu John, David J Brady, Rebecca M Willett, and Timothy J Schulz. Single-shot compressive spectral imaging with a dual-disperser architecture. *Optics express*, 15(21):14013–14027, 2007. [3](#)
- [13] Mayank Goel, Eric Whitmire, Alex Mariakakis, T Scott Saponas, Neel Joshi, Dan Morris, Brian Guenter, Marcel Gavrilu, Gaetano Borriello, and Shwetak N Patel. Hypercam: hyperspectral imaging for ubiquitous computing applications. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, pages 145–156, 2015. [2](#)
- [14] Rahul Gulve, Navid Sarhangnejad, Gairik Dutta, Motasem Sakr, Don Nguyen, Roberto Rangel, Wenzheng Chen, Zhengfan Xia, Mian Wei, Nikita Gusev, et al. 39 000-subexposures/s dual-adc cmos image sensor with dual-tap coded-exposure pixels for single-shot hdr and 3-d computational imaging. *IEEE Journal of Solid-State Circuits*, 58(11):3150–3163, 2023. [2](#), [3](#)
- [15] Neelam Gupta. Hyperspectral imager development at army research laboratory. In *Infrared technology and applications XXXIV*, pages 573–582. SPIE, 2008. [2](#)
- [16] Yasunobu Hitomi, Jinwei Gu, Mohit Gupta, Tomoo Mitsunaga, and Shree K Nayar. Video from a single coded exposure photograph using a learned over-complete dictionary. In *2011 International Conference on Computer Vision*, pages 287–294. IEEE, 2011. [3](#)
- [17] Zhewei Huang, Tianyuan Zhang, Wen Heng, Boxin Shi, and Shuchang Zhou. Real-time intermediate flow estimation for video frame interpolation. In *Computer Vision – ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XIV*, page 624–642, Berlin, Heidelberg, 2022. Springer-Verlag. [6](#), [3](#)
- [18] Zhewei Huang, Tianyuan Zhang, Wen Heng, Boxin Shi, and Shuchang Zhou. Real-time intermediate flow estimation

- for video frame interpolation. <https://github.com/hzwer/ECCV2022-RIFE>, 2022. 3
- [19] Jubin Kang, Yongjae Park, Jung-Hye Hwang, Kieop Hong, Insang Son, Jung-Hoon Chun, Jaehyuk Choi, and Seong-Jin Kim. An indirect time-of-flight sensor with tetra-pixel architecture calibrating tap mismatch in a single frame. *IEEE Solid-State Circuits Letters*, 5:284–287, 2022. 3
 - [20] Chloe LeGendre, Xueming Yu, Dai Liu, Jay Busch, Andrew Jones, Sumanta Pattanaik, and Paul Debevec. Practical multispectral lighting reproduction. *ACM Transactions on Graphics (TOG)*, 35(4):1–11, 2016. 1
 - [21] Chunyu Li, Yusuke Monno, Hironori Hidaka, and Masatoshi Okutomi. Pro-cam ssfm: Projector-camera system for structure and spectral reflectance from motion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2414–2423, 2019. 2
 - [22] Daoyu Li, Jinxuan Wu, Jiajun Zhao, Hanwen Xu, and Liheng Bian. Spectratrack: megapixel, hundred-fps, and thousand-channel hyperspectral imaging. *Nature Communications*, 15(1):9459, 2024. 3
 - [23] Lingen Li, Lizhi Wang, Weitao Song, Lei Zhang, Zhiwei Xiong, and Hua Huang. Quantization-aware deep optics for diffractive snapshot hyperspectral imaging. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19748–19757, 2022. 3, 6, 7, 4, 5
 - [24] Yuqi Li, Qiang Fu, and Wolfgang Heidrich. Dataset for multispectral illumination estimation using deep unrolling network, 2021. 6, 4
 - [25] Dengyu Liu, Jinwei Gu, Yasunobu Hitomi, Mohit Gupta, Tomoo Mitsunaga, and Shree K Nayar. Efficient space-time sampling with pixel-wise coded exposure for high-speed imaging. *IEEE transactions on pattern analysis and machine intelligence*, 36(2):248–260, 2013. 2, 3
 - [26] Patrick Llull, Xuejun Liao, Xin Yuan, Jianbo Yang, David Kittle, Lawrence Carin, Guillermo Sapiro, and David J Brady. Coded aperture compressive temporal imaging. *Optics express*, 21(9):10526–10545, 2013. 3
 - [27] Guolan Lu and Baowei Fei. Medical hyperspectral imaging: a review. *Journal of biomedical optics*, 19(1):010901–010901, 2014. 1
 - [28] Yi Luo, Derek Ho, and Shahriar Mirabbasi. Exposure-programmable cmos pixel with selective charge storage and code memory for computational imaging. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 65(5):1555–1566, 2017. 2, 3
 - [29] Maksim Makarenko, Arturo Burguete-Lopez, Qizhou Wang, Fedor Getman, Silvio Giancola, Bernard Ghanem, and Andrea Fratalocchi. Real-time hyperspectral imaging in hardware via trained metasurface encoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12692–12702, 2022. 3
 - [30] Julien NP Martel, Lorenz K Mueller, Stephen J Carey, Piotr Dudek, and Gordon Wetzstein. Neural sensors: Learning pixel exposures for hdr imaging and video compressive sensing with programmable sensors. *IEEE transactions on pattern analysis and machine intelligence*, 42(7):1642–1653, 2020. 3
 - [31] Ben Niu, Weilei Wen, Wenqi Ren, Xiangde Zhang, Lianping Yang, Shuzhen Wang, Kaihao Zhang, Xiaochun Cao, and Haifeng Shen. Single image super-resolution via a holistic attention network. In *Computer Vision – ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XII*, page 191–207, Berlin, Heidelberg, 2020. Springer-Verlag. 6, 7, 3
 - [32] Jong-Il Park, Moon-Hyun Lee, Michael D Grossberg, and Shree K Nayar. Multispectral imaging using multiplexed illumination. In *2007 IEEE 11th International Conference on Computer Vision*, pages 1–8. IEEE, 2007. 2
 - [33] Wallace M Porter and Harry T Enmark. A system overview of the airborne visible/infrared imaging spectrometer (aviris). In *Imaging spectroscopy II*, pages 22–31. SPIE, 1987. 2
 - [34] Ramesh Raskar, Amit Agrawal, and Jack Tumblin. Coded exposure photography: motion deblurring using fluttered shutter. In *Acm Siggraph 2006 Papers*, pages 795–804. 2006. 3
 - [35] Zheng Shi, Xiong Dun, Haoyu Wei, Siyu Dong, Zhanshan Wang, Xinbin Cheng, Felix Heide, and Yifan Peng. Learned multi-aperture color-coded optics for snapshot hyperspectral imaging. *ACM Transactions on Graphics (TOG)*, 43(6):1–11, 2024. 3
 - [36] Suhyun Shin, Seokjun Choi, Felix Heide, and Seung-Hwan Baek. Dispersed structured light for hyperspectral 3d imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24997–25006, 2024. 2
 - [37] Suhyun Shin, Seungwoo Yoon, Ryota Maeda, and Seung-Hwan Baek. Dense dispersed structured light for hyperspectral 3d imaging of dynamic scenes. *arXiv preprint arXiv:2412.01140*, 2024. 2
 - [38] Edwin Vargas, Julien NP Martel, Gordon Wetzstein, and Henry Arguello. Time-multiplexed coded aperture imaging: Learned coded aperture and pixel exposures for compressive imaging systems. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2692–2702, 2021. 2
 - [39] Dhruv Verma, Ian Ruffolo, David B Lindell, Kiriakos N Kutulakos, and Alex Mariakakis. Chromaflash: Snapshot hyperspectral imaging using rolling shutter cameras. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(3):1–31, 2024. 2
 - [40] Ashwin Wagadarikar, Renu John, Rebecca Willett, and David Brady. Single disperser design for coded aperture snapshot spectral imaging. *Applied optics*, 47(10):B44–B51, 2008. 3
 - [41] Mian Wei, Navid Sarhangnejad, Zhengfan Xia, Nikita Gusev, Nikola Katic, Roman Genov, and Kiriakos N Kutulakos. Coded two-bucket cameras for computer vision. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 54–71, 2018. 2, 3
 - [42] Zhiyang Yao, Shuyang Liu, Xiaoyun Yuan, and Lu Fang. Specat: Spatial-spectral cumulative-attention transformer for high-resolution hyperspectral image reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer*

- Vision and Pattern Recognition (CVPR)*, pages 25368–25377, 2024. [6](#), [7](#)
- [43] Fumihito Yasuma, Tomoo Mitsunaga, Daisuke Iso, and Shree K Nayar. Generalized assorted pixel camera: postcapture control of resolution, dynamic range, and spectrum. *IEEE transactions on image processing*, 19(9): 2241–2253, 2010. [6](#), [4](#)
 - [44] Seung Chul Yoon, Bosoon Park, Kurt C Lawrence, William R Windham, and Gerald W Heitschmidt. Line-scan hyperspectral imaging system for real-time inspection of poultry carcasses with fecal material and ingesta. *Computers and Electronics in Agriculture*, 79(2):159–168, 2011. [1](#)
 - [45] Bohan Yu, Jinxiu Liang, Zhuofeng Wang, Bin Fan, Art Subpa-asa, Boxin Shi, and Imari Sato. Active hyperspectral imaging using an event camera. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 929–939, 2025. [2](#)
 - [46] Jie Zhang, Tao Xiong, Trac Tran, Sang Chin, and Ralph Etienne-Cummings. Compact all-cmos spatiotemporal compressive sensing video camera with pixel-wise coded exposure. *Optics express*, 24(8):9013–9024, 2016. [2](#)
 - [47] Yuanyuan Zhao, Hui Guo, Zhan Ma, Xun Cao, Tao Yue, and Xuemei Hu. Hyperspectral imaging with random printed mask. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10149–10157, 2019. [3](#)
 - [48] Tiancheng Zhi, Bernardo R Pires, Martial Hebert, and Srinivasa G Narasimhan. Multispectral imaging for fine-grained recognition of powders on complex backgrounds. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8699–8708, 2019. [1](#)